

Chipset Integration:

Engineers today are looking for MCU-integrated modules

Technologies such as the Internet of Things (IoT) are providing unparalleled opportunities for businesses. New modules with rising levels of integration are easily available and cost-effective. So, how does one decide between a module and a chipset for an application? Let us find out as Abhishek Mutha of EFY speaks with Dhiraj Sogani, GM and senior VP, Systems Business Unit at Redpine Signals Inc.



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Q. Most IoT devices are designed to be small in size. What are engineers looking at while selecting a wireless module?

A. If everything is integrated together in a module, the overall size and cost of the IoT solution will also be extremely less, making products competitive. Engineers today are looking for microcontroller (MCU) integrated modules where they can write their own applications.

Integration of software is another factor looked at. Traditionally, an MCU was being used to run all software, but today engineers want a major chunk of software to run on the module itself. Modules not only run software of wireless protocols but also networking software like Transmission Control Protocol/Internet Protocol (TCP/IP) stack, Secure Sockets Layer (SSL)/Transport Layer Security (TLS) or Hypertext Transfer Protocol Secure (HTTPS).

Q. Majority of IoT devices are battery-operated. How can low power consumption be achieved?

A. Peak current consumption and overall average current of a module must be low. Modules have multiple sleep modes available, making it possible to turn protocols and devices on and off as desired, thereby helping in reducing power consumption.

In addition to keeping power consumption low, regulatory certifications [such as Federal Communications Commission (FCC) and Industry Canada certification (IC)] and protocol compliances [such as Wi-Fi Alliance and Bluetooth Special Interest Group (SIG)] are also critical in selecting the right solution as the timeline, cost and risk associated with these is quite significant.

Q. How important is it for a module to provide dual-band support for Wi-Fi technology?

A. Wi-Fi operates in two spectra, namely, 2.4GHz and 5GHz. 2.4GHz band is very crowded today because several other technologies such as Bluetooth are operating in this spectrum. The number of devices operating in this band is increasing significantly and the number of available channels is very less. For quality and reliability of connection, customers have started moving to 5GHz band.

Q. What kind of integration would you recommend for IoT products if cost was the only key factor?

A. Designers are moving towards highly-integrated wireless modules, and chipset level integration is going down.

The capital and expertise needed for wireless chipset integration is enormous. You need to have radio frequency expertise, necessary test tools in place and you need to go through reliability testing, performance testing, regulatory certifications and protocol compliance to ensure that you are meeting all the requirements. This increases the chipset integration cost, driving customers to an integrated module, where all these issues are taken care of.

Q. What do you think are the parameters design engineers should consider before buying modules?

A. The first thing that an engineer needs to look at is data throughput requirements of the application. A set-top box or a wireless display will have completely different requirements as compared to a machine-control application.

Second factor is power consumption. The designer should decide if the device is going to be wall-powered or battery-operated. It is also important to ascertain the maximum peak power consumption that the device can sustain.

If using a Linux operating system (OS), most of the software can be run on the MCU without any problem. But if a small real-time operating system or no OS is used, you will need a module that literally runs everything.

Q. You recently announced a module for vehicle-to-vehicle communication. How is your solution unique?

A. This module uses 802.11p and we are incorporating standard Wi-Fi, Bluetooth and ZigBee into this module to make it a complete vehicular communication solution. It would help in creating onboard units that go inside the car or road-side units that are mounted for collision avoidance, traffic scenario communication between different vehicles, toll collection and several other applications.

Several automobile companies are looking into such solutions. We expect them to go into mass adoption in 2016. We also have solutions coming up where an MCU will be integrated inside the modules. ●